Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

> **Golden rule:** nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py`.

Status legend

	Meaning
✅ **Validated**	Survived friction-aware, regime-split backtest (still preliminary — see caveats)
🟡 **Marginal**	Real effect, but fragile net of costs / not sign-stable
❌ **Failed**	Tested and does not work (kept here so we don't retry it)
🧪 **Untested**	Built and wired, awaiting data to evaluate
💡 **Idea**	Backlog — not yet built
🔄 **In progress**	Data/build task underway

Progress at a glance

Last updated: 2026-06-02. Update this section whenever a status, metric, or next-step changes.

Strategy scoreboard

Strategy	Class	Status	Headline result	Next action
`retail_fear` (capitulation / bear-extreme / pessimism-fade)	Macro overlay	✅ Validated	**excess vs buy-&-hold**, generalises across a risk-on basket: Gold +0.6 / AUDJPY +0.5 (win 88%) / SP500 +0.2pp robust; Silver +2.2pp lumpy; VIX gate ~4×	live via MT5 `RedditMacro_EA`; vol-target sizing; longer history
`rrai_momentum_long`, `froth_high_long`	Macro overlay	❌ No edge	positive net but ~0 **excess vs buy-&-hold** (just bull drift)	dropped
`rrai_euphoria_short` + all macro shorts	Macro overlay	❌ Failed	−1.5 to −3pp excess (shorting the bull)	dropped
`dump_fade_DELETION_short`	Equity	✅ Thesis validated (forensic) / ⚠️ not yet PIT-tradeable	**gone≥0.20 → +14.3%/10d short, 83% win, t=2.57, +ve in all 3 regimes**; control proves the gate flips the trade (un-gated pump short *loses* −10.8%, t=−2.22)	make tradeable: finish author sweep to power the PIT `young_frac` gate (now n=3–7, inconclusive); express via puts
`distribution_short` (concentrated pump + sentiment–price divergence)	Equity	🟠 Promising — **PIT-tradeable**	concentrated pump (per_author≥2, m≥10) + sentiment≥0.4 + 5d price already rolling over → **+7.2%/10d net short, t=2.70**; **date-clustered t=2.52 (47 indep. dates), dose-response in sentiment, +ve all 3 sub-periods, survives 2× costs**; control (same pumps, no divergence) *loses* −4.1%	OOS-collect forward; tighten `per_author` (some ETF/large-cap leak in); express via puts; promote on more episodes
`fade_organic_short`	Equity	🟠 Marginal	+0.83%/5d but P3 flips	retry via puts + VIX/vol gate
`pump_long` (ride)	Equity	❌ Failed	+3% but regime-flips, t=0.75	— drop
`dump_fade_short` (naive)	Equity	❌ Failed	−9.9% (squeezed)	superseded by deletion-gated
`organic_long`	Equity	❌ Failed	−4.2%, sign-stable loser	inverse → `fade_organic_short`

Build & data progress

Component	Status	Detail
Feature store (`feature_daily` / `aggregate_daily`)	✅ Done	~77k ticker-day rows, point-in-time
Friction-aware backtester (`backtest/engine.py`)	✅ Done	costs + regime split + no-lookahead
`/signals` feed (multi-strategy, json|mt5)	✅ Done	strategy-tagged; live, currently **flat**
Macro strategy R&D (`backtest/macro_research.py`)	✅ Done	only `retail_fear` beats buy-&-hold; momentum/froth no edge
MT5 framework (`SignalClient.mqh` + `RedditMacro_EA.mq5`)	✅ Done	strategy-selectable thin client; demo-only
Author profiling coverage	🔄 20%	30,015 / 152,040; 2,973 gone (2,087 del / 886 susp); target ~40%
Deletion-gated short testability	✅ Forensically validated	gone≥0.20 short +14.3%/10d, t=2.57 (n=30); PIT `young_frac` proxy still sparse (n=3–7)
Short-interest / squeeze data (FINRA Reg SHO)	✅ Done	`scraper/finra.py` : 475k rows / 5,693 tickers; **z-score** screen (abs level is MM noise, median 0.49)
Independent fear-confirmation gate (Wikipedia)	🟡 Plausible but UNPROVEN	**Committed excess study (`backtest/fear_gate.py`) settles it:** fear-only *reproduces* the +0.36pp point estimate (SPY +0.32 / QQQ +0.41 excess) — so not fabricated — but it's **not significant** (excess t≈0.6, n≈94). The LIVE **trend+fear** combo is *worse* (dilutes SPY→+0.09, flips QQQ→**−0.20**). MT5 absolute showed losses (COVID-cluster). Net: **fear-only > trend+fear for excess**; gate stays thin/unproven; AUDJPY rightly ungated
Live screens in `/signals` (`tradeable:false`)	✅ Done	`smallcap_pump_fade` · `fresh_pump_alert` · `coordination_flag` · `biotech_catalyst` · `attention_fade` · `calendar_overlay`
Idea backlog	💡 7 ideas	squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination
Equity/options paper client (Alpaca)	💡 Planned	not started

Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data ≤ D; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Index / risk-FX** — *macro overlay only* (retail positioning as a contrarian factor).
- **FX majors / "Reddit sentiment on SPY as direction"** — no edge; don't build it.

1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

> **Macro R&D verdict (`backtest/macro_research.py`, excess over buy-and-hold).** A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~0 excess (pure drift); all shorts were −1.5 to −3pp (fighting the trend). **One real macro edge, three formulations of it.**

✅ `retail_fear` — buy risk when retail capitulates *(served live)*

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct ≤ 0.15` **AND** `VIX ≥ 18`; hold ~10d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate ~4×'d it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally → size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- 🟡 **Independent fear-confirmation gate (Wikipedia, LIVE — but now DOUBTFUL):** the **equity** legs (US500/USTEC) additionally require an *independent* fear spike — `fear_z ≥ 0.5`, the z-score of Wikipedia fear-page attention (Stock_market_crash / Recession / Bear_market views) vs the trailing ~30d — **AND** a 200-DMA uptrend. Original rationale: SPY capitulation *alone* doesn't beat buy-and-hold (master backtest excess −0.05); an *ad-hoc* run reported **+0.36pp/10d when `fear_z` confirms vs ~−0.01 ungated**, and fear-attention is **1.32× higher on capitulation days** (`DATA_PLAN.md`). Wired live in `analytics/signals.py` (`_fear_z`, `FEAR_MIN`; `fear_z` in `/signals`).

- 🔬 **MT5 Strategy-Tester A/B (US500, D1, 2020–2026, long-only, real spread/swap; `mql5/RedditMacro_US500_*.ini`, dates via `python main.py --export-signals`).** This tests the gate as an **absolute long-only rule** (NOT excess-vs-buy-&-hold):

variant	signals	trades	win%	net $	PF	maxDD%
baseline (all caps)	335	100	57.0	**+12**	1.00	13.6
`fear_z≥0.5` only	95	45	53.3	**−1045**	0.68	13.0
trend up (200-DMA) only	223	71	56.3	**+21**	1.01	9.0
trend **AND** fear (the live gate)	54	27	55.6	**−372**	0.73	7.1

  Result: the `fear_z` gate does NOT help — it hurts. Adding fear makes US500 *worse* both alone (−$1045) and with trend (−$372 vs trend-only +$21). Fear spikes cluster *at/just before tops* where the 200-DMA still reads up (e.g. Feb-2020) → `trend AND fear` fires **long into the COVID falling knife**. The **trend filter** is the only component that helps, and mostly by cutting drawdown (13.6→7.1%), not return. Baseline ≈ breakeven re-confirms equity capitulation has no absolute edge. - ✅ **Committed EXCESS study (`backtest/fear_gate.py`, `python -m backtest.fear_gate`) — this settles it.** The frame MT5 can't measure: mean EXCESS over buy-&-hold (the +0.36pp's own metric). Same point-in-time date-sets as the MT5 run (shared `capitulation_sets()`). 10d, net of costs:

leg	variant	n	excess%	win%	t(exc)
SPY	baseline	332	−0.05	61	−0.21
SPY	**fear**	94	**+0.32**	68	+0.64
SPY	trend	220	−0.03	60	−0.16
SPY	trend+fear *(live)*	53	+0.09	64	+0.19
QQQ	baseline	332	+0.06	62	+0.20
QQQ	**fear**	94	**+0.41**	65	+0.71
QQQ	trend+fear *(live)*	53	**−0.20**	60	−0.33
AUDJPY *(ungated)*	baseline	333	**+0.26**	68	**+2.04**

Verdict (three honest parts): (1) **The +0.36pp REPRODUCES as a point estimate** — fear-only SPY +0.32 / QQQ +0.41 excess — so it was *not* fabricated; the fear gate does lift equity-capitulation excess. (2) **But it is NOT statistically significant** (excess t≈0.6, n≈94) — a thin, unproven edge. (3) **The LIVE `trend+fear` combo is the wrong way to apply it**: the trend filter *removes the buy-fear-in-a-downtrend bounces that carry the excess*, diluting SPY (+0.32→+0.09) and flipping QQQ negative (+0.41→**−0.20**). - **Reconciliation (why MT5 said "loss" and this says "small win"):** MT5 sums *absolute* P&L and holds one position at a time, so a few COVID-cluster trades dominate the dollar total (−$1045); the excess frame means-averages independent trades, where fear is mildly +ve. Both are correct for their question — for *signal research* the excess frame is the right one, so the earlier "fear-gate is a drag" was a frame artifact. - **The genuine tension:** fear → better *timing/excess* but worse *drawdown* (COVID knives); trend → better *drawdown* but worse *excess*. The live `trend AND fear` gets the worst of both for excess. - **Recommendation:** keep the gate 🟡 (positive but insignificant). If we ever optimise the equity legs for excess, use **fear-only, not trend+fear**; AUDJPY (the one significant edge, +0.26 t=2.04) stays **ungated** — confirmed correct. (User chose to leave the live gate unchanged for now; this is logged for when we revisit.) - **Expression:** a **diversified risk-on basket** — AUDJPY + Gold (XAUUSD) + US500/USTEC + small Silver (XAGUSD). Served via `/signals` (tag `retail_fear`); traded by `mql5/RedditMacro_EA.mq5` (demo). Equity legs gated by trend + `fear_z`. - **Caveats:** single ~12mo bull regime; **46 cap-days = only ~13 distinct fear episodes** (clustered → effective n small); overlapping windows; edge small in absolute terms. - **Next:** leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

❌ Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short`, `bull_extreme_short`, `froth_high_short`, all momentum-shorts: **−1.5 to −3pp excess** — shorting a bull market.
- `rrai_up_momentum_long`, `froth_high_long`: **~0 excess** — just drift, no signal.

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2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

❌ `pump_long` — ride concentrated promotion

- **Thesis:** concentrated mention spike (few authors, high volume) pops short-term.
- **Result:** net +3%/5d but **sign-flips** (P1 +15.8 → P3 −4.6), win 36%, t=0.75. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust.**

❌ `dump_fade_short` — short a decelerating pump (naive)

- **Result:** net **−9.9%** (P1 −29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

❌ `organic_long` — buy broad attention spikes

- **Result:** sign-stable **−4.2%** net (t=−4.25, n=321). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

🟡 `fade_organic_short` — fade broad attention spikes

- **Thesis:** broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result:** net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal.** Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

✅ `dump_fade_DELETION_short` — short a pump whose promoters vanished *(THE thesis — forensically validated)*

- **Thesis:** concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle.
- **Result (2026-06-02, 20% author coverage, h=10, market-neutral, 75bps + 8bps/day borrow):** | gate | n | net short | win | t | regimes | |---|---|---|---|---|---| | `gone_frac≥0.10` | 33 | **+10.0%** | 81.8% | 1.51 | −1.2 / +19.9 / +11.4 | | `gone_frac≥0.20` | 30 | **+14.3%** | 83.3% | **2.57** | +11.4 / +27.7 / +3.9 | | **control:** pumps w/ promoters *still here* (`gone=0`) | 202 | **−10.8%** | 36.1% | −2.22 | −30.7 / −0.6 / −1.1 |
- The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at gone≥0.20.
- ⚠️ **Not yet point-in-time tradeable.** `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude (n=3–7, and the few events lean the wrong way).
- **Path to tradeable:** (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

🟡 `distribution_short` — concentrated pump + sentiment-price divergence *(NEW, point-in-time tradeable)*

- **Thesis:** the "bagholder distribution" moment — a *manufactured* pump (concentrated chatter) where **price has started rolling over but the crowd is still euphoric**. Insiders distribute into the retail bid; sentiment lags price. Every input (`per_author` , `mentions` , `sentiment` , trailing price return) is known at signal time → **unlike the deletion gate, this is actually tradeable live.** (`backtest/microstructure.py`)
- **Rule:** `per_author≥2 AND mentions≥10 AND sentiment≥0.4 AND 5d-trailing-return<0` → SHORT. h=10, market-neutral, 75bps + 8bps/day borrow.
- **Result (2026-06-02): +7.2%/10d net, t=2.70, win 52%, n=60** across **39 distinct tickers**.
- **Date-clustered test** (1 obs/date, kills intra-day correlation): **+7.6%, t=2.52, 47 independent dates** — *not* a risk-off-day artifact (max 3 events on any date, spread over 12 months).
- **Dose-response** in sentiment (s≥0.3/0.4/0.5 → +3.9/+7.2/+11.1%) and **monotone in horizon** (h=5/10/20 → +5.6/+7.2/+9.5%, t rising to 2.92) — the hallmarks of a real effect, not a fit.
- **Survives 2× costs** (150bps spread + 16bps/day borrow → still +4.9%/10d).
- **Control:** same concentrated pumps *without* the divergence (price up) → **−4.1%** — the divergence is doing the work. Positive in all 3 sub-periods.
- **Why it works where the others don't:** the gate is *precise* — it only fires inside the manufactured subset. The broad high-attention universe **drifts flat-to-up** (gross ≈ +0.04%), so generic pump-shorting just pays costs (see negatives below); the edge requires isolating the distributing pumps, which both this and the deletion gate do.
- **Caveats / next:** n still modest (47 indep. dates); single ~12-mo window (sub-periods, not true bull/bear regimes); small-cap-heavy so no multi-regime price history; loose `per_author≥2` lets some ETFs/large-caps (QQQ/IWM) leak in — tighten to ≥3 and/or split small-cap vs index. **Collect forward OOS** (the clock starts now); express via **puts**. Not yet promoted to the live `/signals` feed.

❌ Negatives from the same study (recorded so we don't re-run them)

- **Broad-universe microstructure timing does NOT survive costs.** Recruitment-exhaustion, mention-deceleration, and sentiment-price divergence applied to the *whole* high-attention set (mentions_z≥1) are all **net-negative** — the universe drifts flat-to-up, so shorting it generically just bleeds the ~2.3% friction. Precision (concentration filter) is mandatory.
- **Recruitment-exhaustion alone = null.** New-author inflow rolling over is directionally supportive *within* concentrated pumps (+1.8% net, but t=0.33) and adds to the stack, but is not a standalone edge at current coverage. The new-author feature is built and kept.
- **Up-leg longs both fail.** (a) "Ride while recruiting + accelerating" (concentrated) netted +1.6% but win 27%, regimes [+21/−7/−9] — one big winner, otherwise negative. (b) "Attention-leads-price coil" (broad: recruitment surging while price still flat) was **net −1.4 to −1.9%** across horizons, sign-stable loser. Neither is a tradeable long; dropped.

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3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) **+ short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*
- **Options-flow signal:** parse contract-level chatter (unusual_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

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4. Data gaps blocking strategies

Gap	Unblocks
Intraday prices (have EOD only)	timing pump/dump entries/exits (they move intraday)
Short interest / float / borrow fee	squeeze *level* (FINRA daily short-*volume* now in via `scraper/finra.py` , but it's MM noise at the level; need bi-monthly SI%/float + borrow)
Options chains / IV	options expression; options-flow signal
Longer history (data-limited to ~12mo)	true multi-regime validation of the RRAI overlay
Author coverage (20% → target ~40%+)	a *point-in-time tradeable* deletion short — thesis already forensically validated; PIT `young_frac` proxy needs more coverage

5. Execution stack

- `/signals` **feed** (`analytics/signals.py` , `GET /signals` json\|mt5) — live, serves the capitulation overlay. ✅
- **MT5 EA** (`mql5/RedditRRAI_EA.mq5`) — thin client for the index/FX overlay; demo-only. ✅
- **Alpaca / IBKR paper client** — for the equity/options legs. 💡 planned.

★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, **including the 2022 bear** (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** (`backtest/sectors.py`). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (−14.6%/10d excess, win 22%, **t=−3.93** — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%, t=2.23 — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.
- **Pump detection is TOO SLOW** (`feature_daily`). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total≥20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix:** a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX):** turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (−0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building (`au_aggregate_daily` , post-level) to test AU-sentiment → AUDJPY.

★ MASTER BACKTEST vs BUY-AND-HOLD (`backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---|---| | retail_fear **AUDJPY** | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail_fear SPY | 338 | +0.54 | +0.59 | −0.05 | 61% | **no edge** | | retail_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria_short SPY | 351 | −1.26 | +0.59 | −1.84 | 27% | no edge | | smallcap_pump_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention_fade | 252 | −0.73 | +0.59 | −1.31 | 40% | no edge |

Recalibration: with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

DATA WE NEED (to validate/strengthen — the binding constraints)

Data	Unblocks	Why we lack it
Intraday prices (we have EOD only)	pump entry/exit timing; fresh-pump fade execution (pumps move intraday)	not collected
Historical penny-stock prices + per-ticker features	multi-regime validation of smallcap_pump_fade & biotech (only 2025-26 exists)	delisted pennies → Yahoo gaps; per-ticker feature_daily only recent
Short interest / float / borrow fee	squeeze setups; realistic short cost/feasibility	FINRA daily short-*volume* in (`scraper/finra.py`); still need SI%/float/borrow (Ortex / FINRA bi-monthly SI)
Options chains / IV	express fades via PUTS (only sane way to short pennies); options-flow signal	not collected
More years of RRAI history	tighten the macro edge (effective n ≈ a few dozen fear episodes)	Reddit/arctic-shift depth ~2017+
Broader author coverage (~20% profiled)	a *PIT-tradeable* deletion/young-account short (thesis already validated forensically; `young_frac` proxy starved)	rate-limited Reddit /about sweep
Independent sentiment sources (StockTwits, news, Discord, crypto + AU subs)	independent episodes → real effective-sample growth without waiting calendar time	not integrated (AU track building)
Real transaction-cost data (per-instrument spreads/borrow)	trustworthy net returns (small-cap costs are modelled/optimistic)	not sourced

Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

Changelog

- **2026-06-02** — **Committed EXCESS study settles the fear-gate (`backtest/fear_gate.py`).** The proper frame (mean excess vs buy-&-hold, the +0.36pp's own metric). Three findings: (1) the +0.36pp **reproduces** as a point estimate (SPY +0.32 / QQQ +0.41 fear-only excess) — not fabricated; (2) it's **not significant** (excess t≈0.6, n≈94); (3) the LIVE **trend+fear** combo is *worse* — dilutes SPY (→+0.09), flips QQQ negative (→−0.20) — because the trend filter strips the downtrend bounces that carry the excess. Reconciles the MT5 "loss": MT5 sums absolute P&L (COVID-cluster dominates), excess means-averages — so the earlier "drag" was a frame artifact. Gate stays 🟡; fear-only > trend+fear for excess; AUDJPY stays ungated (the only significant edge). Exporter refactored to share the study's `capitulation_sets()` (MT5 + Python test identical dates).
- **2026-06-02** — **MT5 backtest of the fear-gate → it's a DRAG, not a rescue.** Extended `--export-signals` to emit 4 point-in-time capitulation date-sets (baseline / `fear_z≥0.5` / 200-DMA-trend / trend+fear) and ran each through the MT5 Strategy Tester on US500 (2020-26) via `RedditMacro_US500_*.ini` . Adding `fear_z` made absolute returns *worse* (alone −$1045, with trend −$372 vs trend-only +$21) — fear spikes fire long into tops (Feb-2020). Only the trend filter helps (DD 13.6→7.1%). Caveat: MT5 = absolute long-only, the +0.36pp was *excess* — so not a direct refutation, but the gate is downgraded to 🟡 doubtful and needs a committed *excess-based* study to settle (maybe drop `fear_z` , keep trend).
- **2026-06-02** — **Doc-completeness pass.** Audited the doc against the live registry + all backtest modules. Found one missing edge — the **Wikipedia fear-confirmation gate** (live: gates the equity legs of `retail_fear` , `fear_z≥0.5` , +0.36pp rescue) — now documented, *honestly flagged as not yet committed-backtested*. Also added the full live-screen list (`fresh_pump_alert` , `coordination_flag` , ...) to the scoreboard and recorded the broad "attention-leads-price" long as a failed up-leg test.

- **2026-06-02** — **New PIT-tradeable short found:** `distribution_short` (`backtest/microstructure.py`). Built the missing **new-author recruitment** feature and tested demand-quality timing signals. The decisive lesson: microstructure timing (recruitment-exhaustion / deceleration / divergence) is **worthless on the broad attention universe** (it drifts flat-to-up, costs dominate) but **powerful inside the concentrated/manufactured subset** — exactly where the deletion edge lives. Winner = concentrated pump + euphoric sentiment + price already rolling over → **+7.2%/10d net, t=2.70 (date-clustered t=2.52, 47 indep. dates), dose-response, survives 2× costs**; control loses −4.1%. Unlike the deletion gate it uses **no hindsight** → tradeable. Recorded negatives (broad timing, recruitment-alone, up-leg ride) to avoid re-runs.
- **2026-06-02** — **Deletion-gated short forensically validated.** Author coverage 8%→20% (2,973 gone). Re-backtest: `gone_frac≥0.20` → +14.3%/10d short (83% win, t=2.57, +ve all regimes); the *control* (un-gated concentrated pumps) *loses* −10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion timestamp) → not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse (n=3–7). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.
- **2026-06-01** — Instrument scan (17 MT5 instruments). `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
- **2026-06-01** — Macro R&D + MT5 framework. Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5` .
- **2026-06-01** — Initial document. Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.